G.V. ACHARYA INSTITUTE OF ENGINEERING & TECHNOLOGY

Department of Computer Science & Engineering

🎓 FINAL YEAR PROJECT REPORT

Vyapar Setu

A Multi-Vendor E-Commerce Marketplace Platform

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Course: B.Tech Computer Science & Engineering

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Abstract

Vyapar Setu is a full-featured multi-vendor e-commerce marketplace platform designed to bridge the gap between independent small and medium enterprise (SME) sellers and consumers across India. Built using **PHP**, **MySQL/SQLite**, and **Bootstrap 5**, the platform enables multiple vendors to register, list products, and manage orders through dedicated dashboards, while customers can browse, search, compare, and purchase products from various sellers through a unified storefront.

The system implements a three-tier role-based architecture supporting **Customers**, **Vendors**, and **Administrators**, each with dedicated portals and permission-based access control. Key features include dynamic product catalog management, shopping cart with AJAX-based real-time updates, wishlist functionality, a complete order lifecycle with status tracking, product review and rating systems, vendor analytics dashboards, and comprehensive admin governance tools.

The platform is built as a server-rendered web application using the LAMP stack (Linux/Windows, Apache, MySQL, PHP) with a responsive front-end powered by Bootstrap 5, Bootstrap Icons, and custom CSS. The database layer supports dual-mode operation with MySQL as the primary database and SQLite as an automatic fallback, ensuring zero-configuration deployment for development and demonstration purposes.

Keywords: Multi-Vendor Marketplace, E-Commerce, PHP, MySQL, Bootstrap, LAMP Stack, Role-Based Access Control, SME, Online Shopping Platform.

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Rishiraj Sharma
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Table of Contents

A. Abstract

B. Acknowledgments

C. Table of Contents

1. Introduction

2. System Requirements

3. Theoretical Background

4. Objectives

5. System Flowchart

6. ER Diagram (Entity-Relationship)

7. Data Flow Diagram (DFD)

8. Algorithm

9. Screenshots & Walkthrough

10. Advantages & Disadvantages

11. Analysis

12. Conclusion

13. References

1 Introduction

1.1 Background

The rapid growth of e-commerce in India, projected to reach $200 billion by 2027, has created a significant demand for digital marketplace platforms that empower small and medium enterprises (SMEs) to sell their products online. Traditional single-vendor e-commerce systems limit product variety and place the entire burden of inventory management, logistics, and customer service on a single entity. Multi-vendor marketplace platforms, such as Amazon, Flipkart, and Etsy, solve this problem by enabling multiple independent sellers to list and sell products through a unified storefront.

Vyapar Setu (meaning "Bridge of Commerce" in Hindi) is a web-based multi-vendor marketplace platform that aims to connect independent artisans, small manufacturers, and verified sellers with consumers. The platform provides dedicated portals for three distinct user roles — Customers, Vendors, and Administrators — enabling a complete e-commerce ecosystem within a single application.

1.2 Problem Statement

Many Indian SMEs and artisans lack the technical expertise and financial resources to build and maintain their own e-commerce websites. Existing marketplace platforms charge high commission rates (15–30%) and impose strict onboarding requirements that exclude small-scale sellers. There is a need for an affordable, easy-to-deploy marketplace platform that enables sellers to set up digital storefronts quickly, manage their inventory, and process orders — all while providing customers with a seamless shopping experience.

1.3 Proposed Solution

Vyapar Setu addresses this gap by providing a self-hosted, open-source multi-vendor marketplace platform that can be deployed on any PHP-compatible web server (such as XAMPP, WAMP, or a cloud VPS). The platform features:

- Role-based authentication with separate dashboards for Customers, Vendors, and Admins
- Complete product lifecycle management with categories, images, pricing, and stock control
- Shopping cart with AJAX-based add/remove operations for seamless user experience
- Wishlist system for saving favorite products
- Complete order processing pipeline with status tracking
- Product review and rating system with vendor response capability
- Vendor analytics dashboard with revenue, sales, and performance metrics
- Admin governance panel with user management, vendor approvals, and platform statistics
- Dual-database support (MySQL primary + SQLite fallback) for zero-configuration deployment

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2 System Requirements

2.1 Hardware Requirements

COMPONENT	MINIMUM SPECIFICATION	RECOMMENDED
Processor	Intel Core i3 / AMD Ryzen 3	Intel Core i5 / AMD Ryzen 5 or higher
RAM	4 GB	8 GB or more
Storage	256 GB HDD	512 GB SSD
Display	1366 × 768 resolution	1920 × 1080 or higher
Network	Internet connectivity (for CDN assets)	Broadband connection

2.2 Software Requirements

SOFTWARE	VERSION	PURPOSE
Operating System	Windows 10/11, Linux, macOS	Host environment
XAMPP / WAMP / LAMP	8.x+	Local server stack
PHP	8.0 or higher	Server-side scripting
MySQL / MariaDB	5.7+ / 10.4+	Primary relational database
SQLite	3.x (bundled with PHP)	Fallback database engine
Apache HTTP Server	2.4+	Web server
Web Browser	Chrome 90+, Firefox 88+, Edge 90+	Client-side rendering
VS Code / Sublime Text	Latest	Code editor (development)

2.3 Technologies & Frameworks Used

TECHNOLOGY	ROLE
PHP 8.x	Backend logic, routing, session management, database operations
MySQL / SQLite (PDO)	Data storage with prepared statements for security
HTML5	Semantic page structure
CSS3 + Custom Stylesheet	Visual theming, responsive layout, animations
Bootstrap 5.3	Responsive grid system, components, utilities
Bootstrap Icons	Iconography throughout the UI
JavaScript (Vanilla)	AJAX cart/wishlist operations, UI interactivity
Chart.js	Analytics charts (vendor & admin dashboards)
AOS (Animate on Scroll)	Scroll-triggered animations on homepage
Google Fonts (Poppins)	Custom typography

3 Theoretical Background

3.1 Multi-Vendor Marketplace Model

A multi-vendor marketplace is an e-commerce platform where products are supplied by multiple third-party sellers while the marketplace operator provides the platform infrastructure. Unlike a single-vendor model (where one business owns all inventory), the marketplace model aggregates products from independent sellers, creating a wider product catalog and competitive pricing. The platform earns revenue through commission fees on each transaction.

The key stakeholders in such a system include: (a) the **Platform Administrator**, who manages the marketplace infrastructure, approves vendors, and monitors operations; (b) **Vendors/Sellers**, who list products, manage inventory, and fulfill orders; and (c) **Customers/Buyers**, who browse, purchase, and review products.

3.2 LAMP Stack Architecture

The LAMP stack (Linux, Apache, MySQL, PHP) is one of the most widely used open-source web development platforms. In this project:

- **Apache** serves as the HTTP web server, handling client requests and routing them to PHP scripts.
- **PHP** processes server-side logic including authentication, database queries, form validation, and page rendering.
- **MySQL/SQLite** stores structured data including user accounts, product catalogs, orders, and reviews via the PDO (PHP Data Objects) abstraction layer.

3.3 Role-Based Access Control (RBAC)

RBAC is a security paradigm that restricts system access based on the roles assigned to individual users. In Vyapar Setu, three roles are implemented:

- **Customer:** Can browse products, manage cart/wishlist, place orders, write reviews, and manage their profile.

- **Vendor:** Can add/edit products, view orders, respond to reviews, manage store settings, and view analytics.
- **Admin:** Can manage all users, approve/reject vendors, manage categories/brands, view platform-wide analytics, and configure settings.

3.4 PDO & Prepared Statements

PHP Data Objects (PDO) provides a uniform interface for accessing multiple databases. The project uses PDO with prepared statements (parameterized queries) to prevent SQL injection attacks. This approach separates SQL code from user input, ensuring that malicious input cannot alter the query structure.

3.5 Session-Based Authentication

The platform uses PHP's native session management for authentication. Upon successful login, a session is created storing the user's ID, name, email, and role. Middleware functions (`requireAuth()`, `requireRole()`) verify session data before granting access to protected pages. Session data is destroyed on logout, and session cookies are invalidated.

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4 Objectives

The primary objectives of the Vyapar Setu project are:

1. **To design and develop a complete multi-vendor e-commerce marketplace** that supports independent product listing, ordering, and fulfillment by multiple sellers through a unified storefront.
2. **To implement a robust role-based access control system** with three distinct user roles (Customer, Vendor, Admin), each with dedicated dashboards and appropriate permissions.
3. **To build a responsive and visually appealing user interface** using Bootstrap 5 that works seamlessly across desktop, tablet, and mobile devices.
4. **To implement secure authentication and authorization mechanisms** using password hashing (bcrypt), session management, and prepared SQL statements to prevent common web vulnerabilities.
5. **To create a complete shopping experience** including product browsing with filters, AJAX-powered shopping cart, wishlist, checkout process, and order tracking.
6. **To provide comprehensive vendor tools** for product management, order fulfillment, customer review management, store settings, and sales analytics.
7. **To develop an administrative governance panel** for platform-wide user management, vendor approval workflows, category/brand management, and financial analytics.
8. **To ensure seamless deployment** by implementing dual-database support (MySQL + SQLite fallback) with automatic schema initialization and seed data generation.

5 System Flowchart

The following flowchart illustrates the complete user journey through the Vyapar Setu platform, from initial access to the order fulfillment cycle.

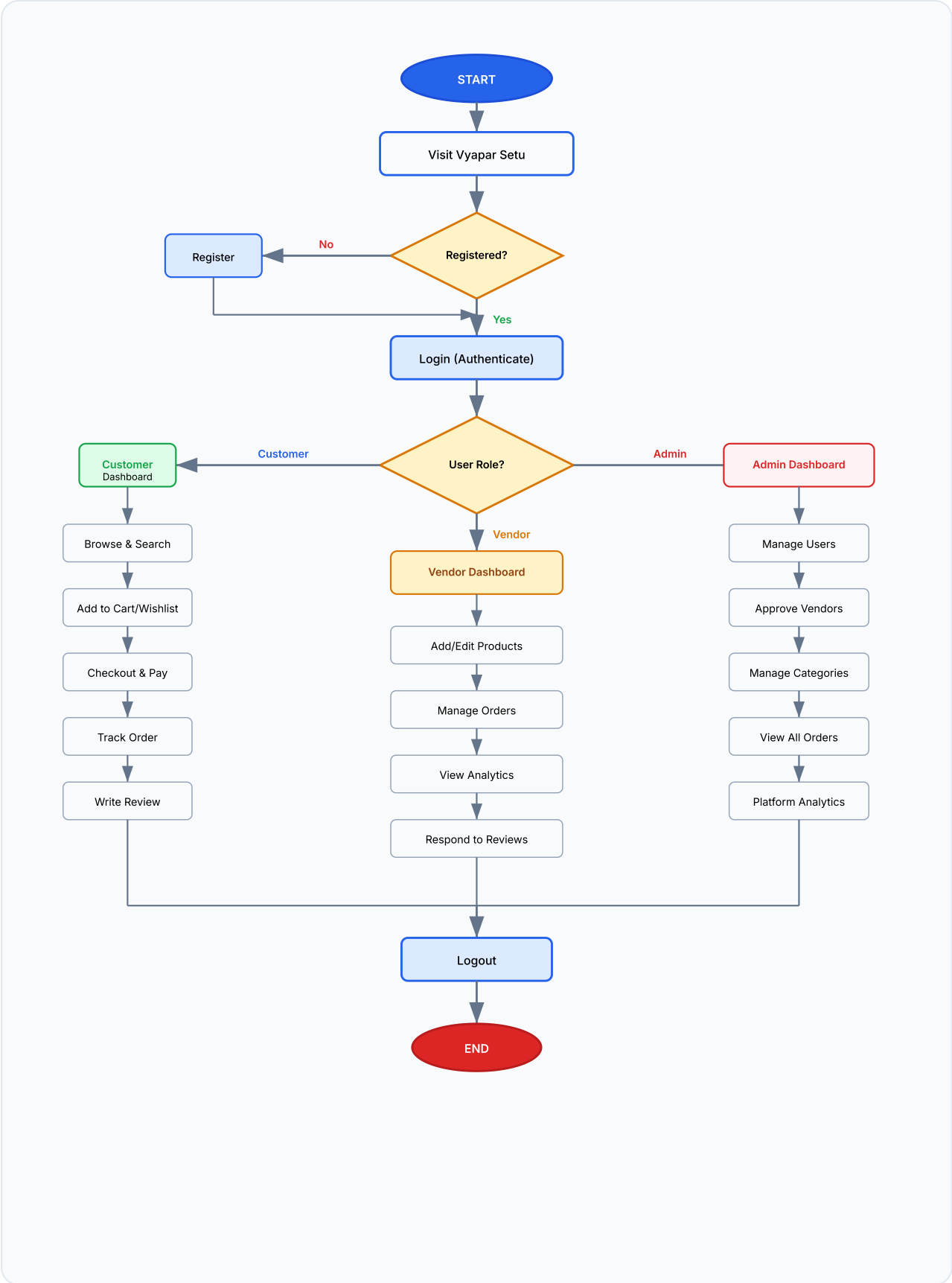


Figure 5.1 — System Flowchart of Vyapar Setu Multi-Vendor Marketplace

6 ER Diagram

The following Entity-Relationship diagram illustrates the database structure of Vyapar Setu, showing all 16 tables and their relationships.

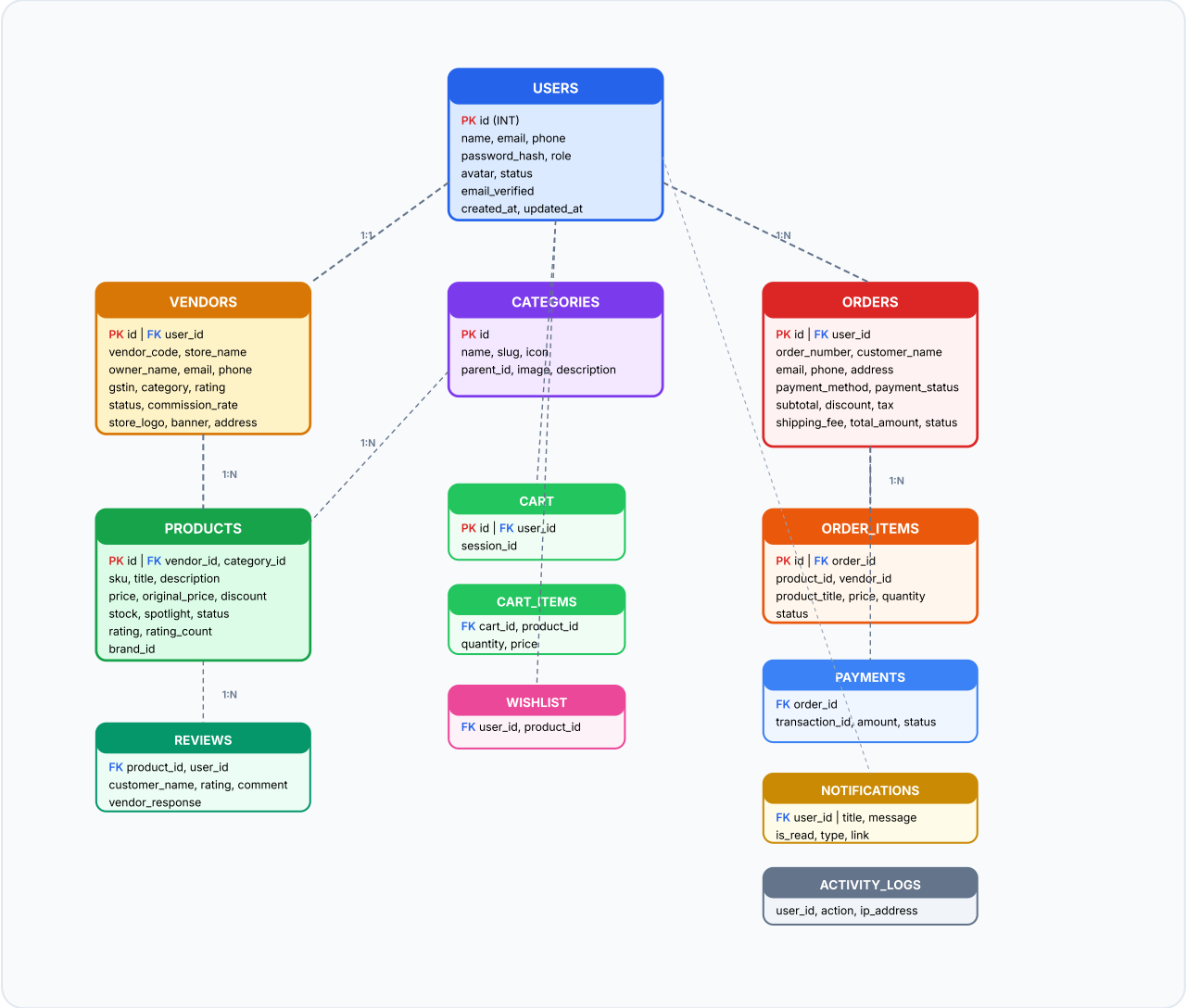


Figure 6.1 — Entity-Relationship Diagram of Vyapar Setu Database (16 Tables)

7 Data Flow Diagram

The Data Flow Diagram (DFD) shows how data moves between external entities, processes, and data stores within the Vyapar Setu system.

7.1 Context Diagram (Level 0 DFD)

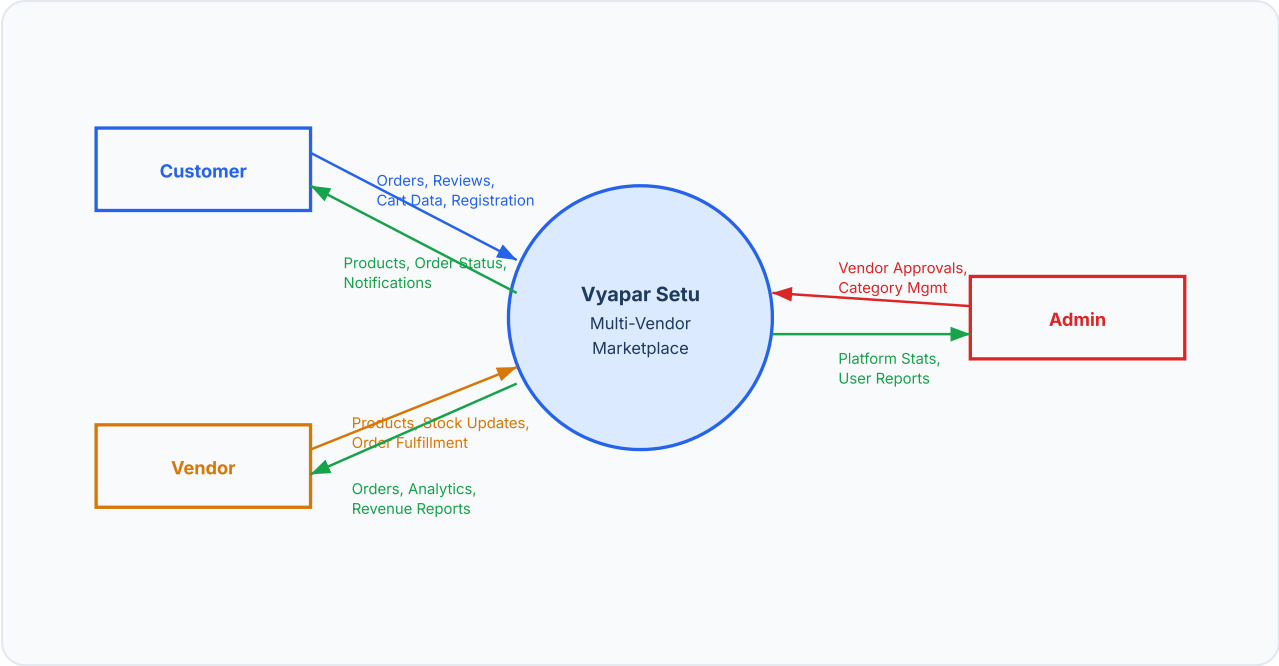


Figure 7.1 — Context Diagram (Level 0 DFD) of Vyapar Setu

7.2 Level 1 DFD — Key Processes

PROCESS	INPUT	OUTPUT	DATA STORE
1.0 User Authentication	Email, Password	Session Token, Role Redirect	D1: Users
2.0 Product Catalog	Category Filters, Search Query	Product Listings, Details	D2: Products, D3: Categories
3.0 Cart Management	Product ID, Quantity	Cart Summary, Updated Count	D4: Cart, D5: Cart Items
4.0 Order Processing	Cart Items, Address, Payment	Order Confirmation, Order Number	D6: Orders, D7: Order Items

PROCESS	INPUT	OUTPUT	DATA STORE
5.0 Product Management	Product Details, Images	Published Listing	D2: Products, D8: Product Images
6.0 Review System	Rating, Comment	Published Review	D9: Reviews
7.0 Admin Governance	Approval Decision, Settings	Status Updates, Reports	D1: Users, D10: Vendors

8 Algorithm

8.1 User Authentication Algorithm

```
Algorithm: USER_LOGIN(email, password)
─────────────────────────────────────
1.  INPUT: email, password from login form
2.  VALIDATE email format and non-empty password
3.  IF validation fails THEN
4.      DISPLAY error message
5.      RETURN to login page
6.  ENDIF
7.  QUERY user record from Users table WHERE email = input_email
8.  IF no user found THEN
9.      DISPLAY "Invalid email or password"
10.     RETURN
11. ENDIF
12. VERIFY password against stored password_hash using bcrypt
13. IF password does NOT match THEN
14.     DISPLAY "Invalid email or password"
15.     RETURN
16. ENDIF
17. IF user.status ≠ 'ACTIVE' THEN
18.     DISPLAY "Account suspended/inactive"
19.     RETURN
20. ENDIF
21. CREATE session with user_id, name, email, role
22. LOG activity: "User logged in" with IP address
23. REDIRECT based on role:
24.     IF role = 'admin'    → Admin Dashboard
25.     IF role = 'vendor'   → Vendor Dashboard
26.     IF role = 'customer' → Customer Dashboard
27. END
```

8.2 Add to Cart Algorithm

```
Algorithm: ADD_TO_CART(product_id, quantity, user_id)
─────────────────────────────────────
1.  INPUT: product_id, quantity (default=1), user_id from session
2.  IF user is NOT logged in THEN
3.      RETURN redirect to login page
4.  ENDIF
```

```
5.  FETCH product price and stock from Products table
6.  IF product NOT found OR stock = 0 THEN
7.      RETURN error "Product unavailable"
8.  ENDIF
9.  FIND cart for user_id in Cart table
10. IF no cart exists THEN
11.     CREATE new cart record for user_id
12.     SET cart_id = newly created ID
13. ELSE
14.     SET cart_id = existing cart ID
15. ENDIF
16. CHECK if product already exists in Cart_Items for this cart_id
17. IF item exists THEN
18.     UPDATE quantity = existing_quantity + new_quantity
19.     UPDATE price = current_product_price
20. ELSE
21.     INSERT new Cart_Item (cart_id, product_id, quantity, price)
22. ENDIF
23. CALCULATE new cart count
24. RETURN success with updated cart_count (JSON response)
25. END
```

8.3 Order Placement Algorithm

```
Algorithm: PLACE_ORDER(user_id, address, payment_method)
_____________________________________

1.  INPUT: shipping details, payment method, user_id
2.  FETCH all cart items for user_id with product details
3.  IF cart is empty THEN
4.      REDIRECT to cart page with warning
5.  ENDIF
6.  CALCULATE subtotal = SUM(price × quantity for each item)
7.  CALCULATE shipping = ₹99.00 (if subtotal > 0)
8.  CALCULATE total = subtotal + shipping
9.  GENERATE unique order_number = "ORD-" + random(10000, 99999)
10. BEGIN database transaction
11. TRY:
12.     INSERT order into Orders table
13.     FOR each cart_item DO:
14.         INSERT into Order_Items (order_id, vendor_id, product details)
15.         UPDATE product stock = MAX(0, stock - quantity)
16.     ENDFOR
17.     DELETE all cart_items for this user
18.     COMMIT transaction
19.     DISPLAY success: "Order placed successfully"
20.     REDIRECT to order confirmation page
21. CATCH exception:
22.     ROLLBACK transaction
```

```
23.     DISPLAY error message
24. END
```

9 Screenshots & Walkthrough

The following screenshots demonstrate the key features and pages of the Vyapar Setu platform.

9.1 Homepage — Hero Banner & Navigation

Homepage — Vyapar Setu Marketplace

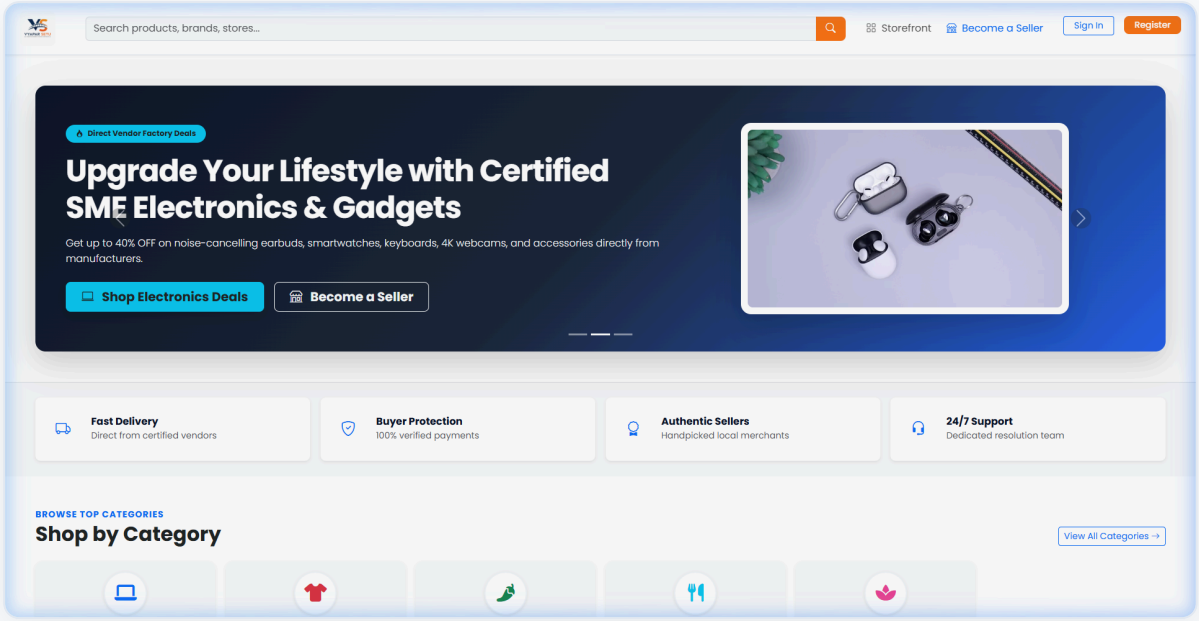


Figure 9.1 — The homepage features a gradient hero banner, global search bar, category browsing, and featured product listings.

9.2 Homepage — Featured Products & Categories

Homepage — Trending Products Section

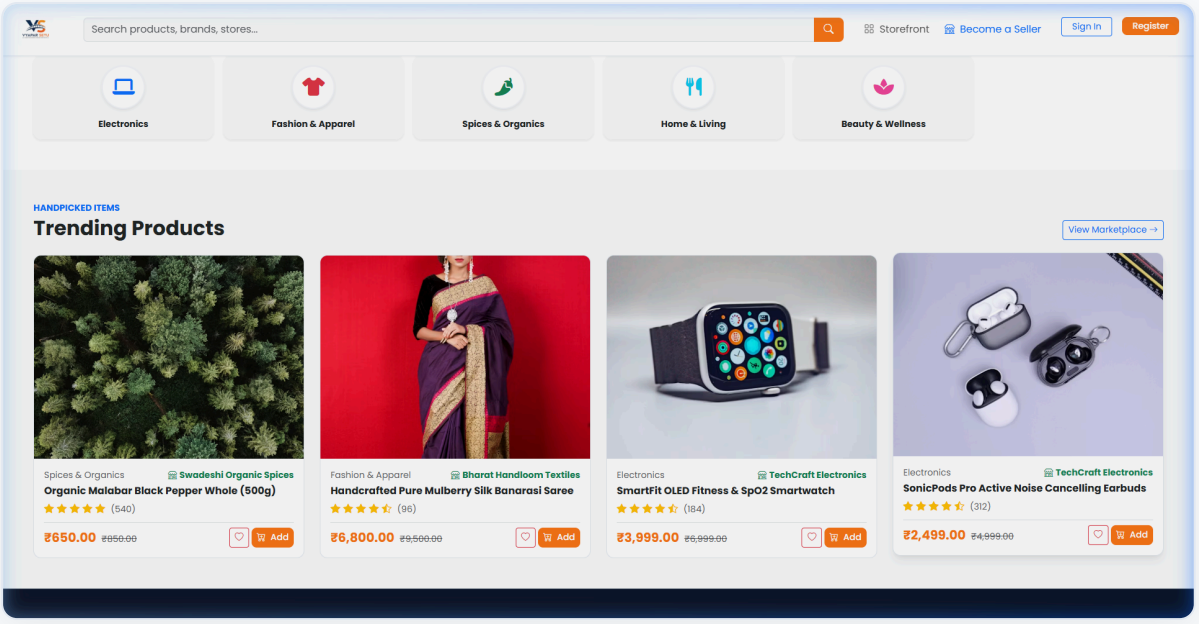


Figure 9.2 — Product cards display pricing, vendor name, ratings, wishlist button, and add-to-cart functionality.

9.3 Shop Marketplace — Product Catalog with Filters

Shop Marketplace — Storefront with Sidebar Filters

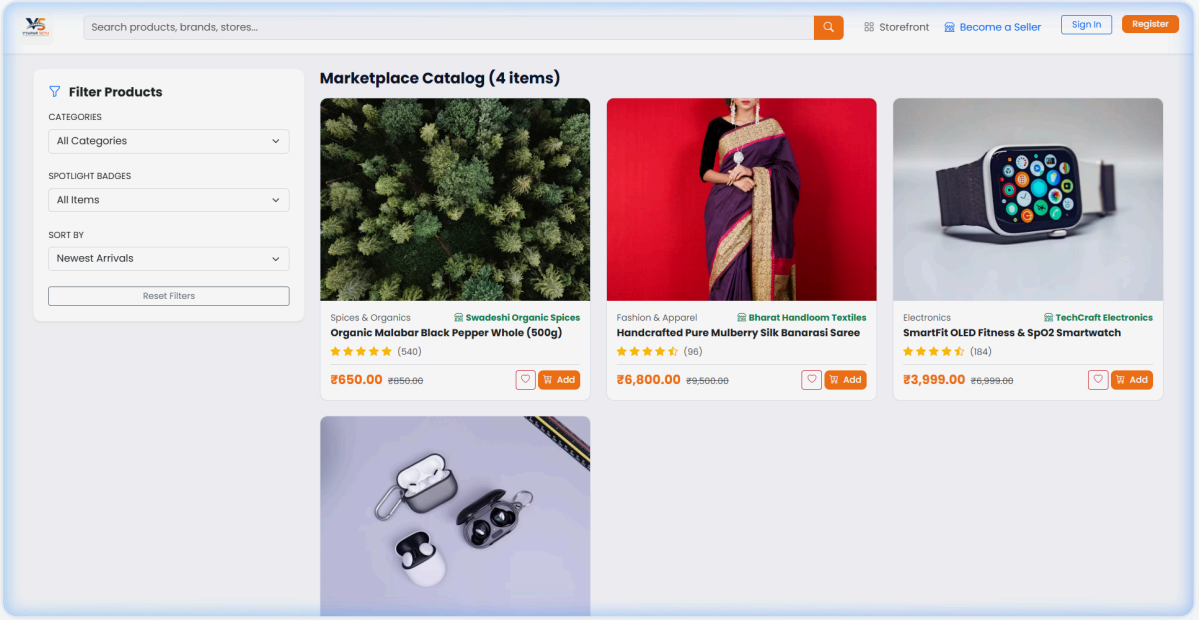


Figure 9.3 — The shop page provides category filters, spotlight badges (Bestseller/Featured), and sort options (newest, price, rating).

9.4 Product Detail Page

Product Detail — SonicPods Pro Earbuds

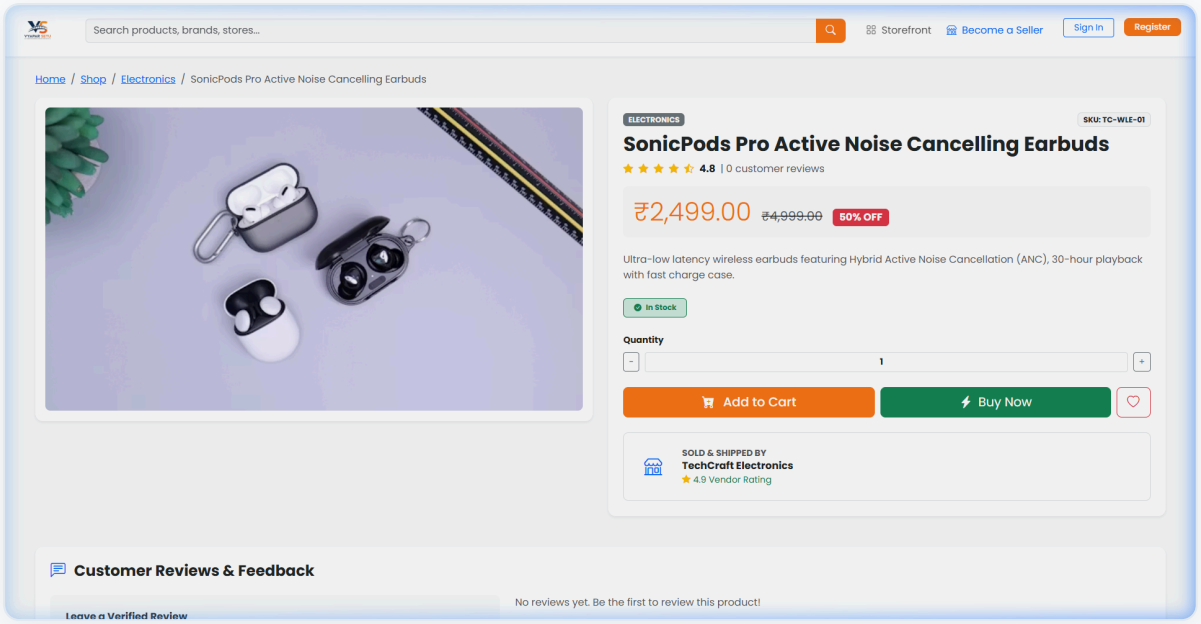


Figure 9.4 — Full product view with image gallery, pricing with discount, stock status, quantity selector, vendor info, and customer reviews section.

9.5 User Login Page

Sign In — Authentication Portal

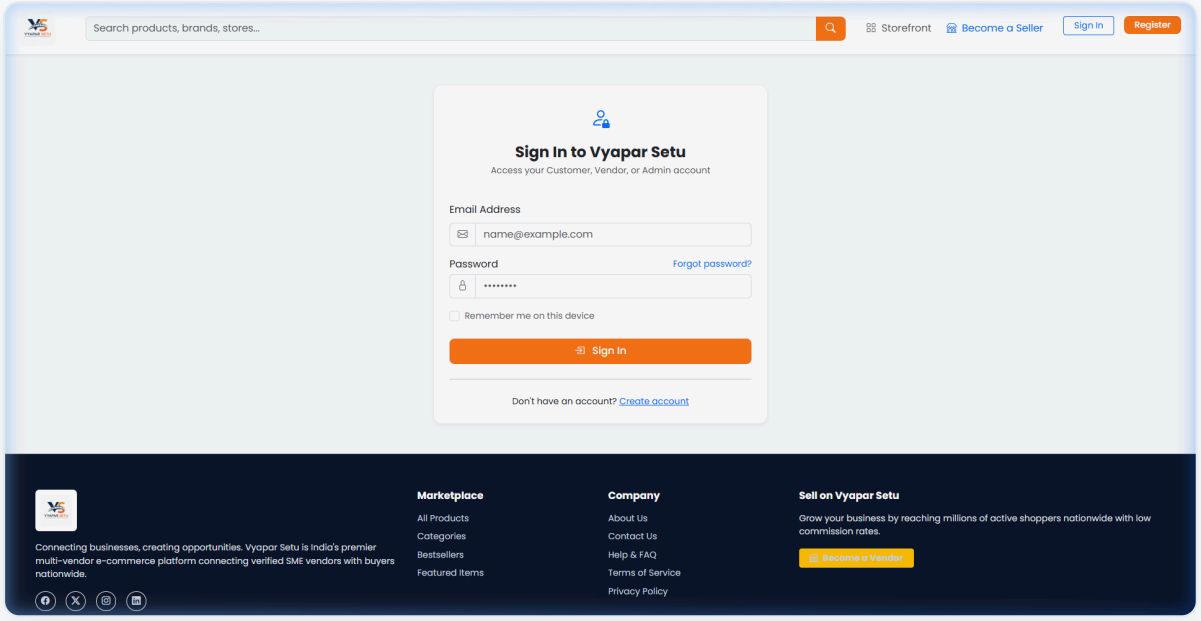


Figure 9.5 — Unified login page for Customers, Vendors, and Admins with role-based redirect after authentication.

9.6 Registration Page

Account Registration — Customer/Vendor Toggle

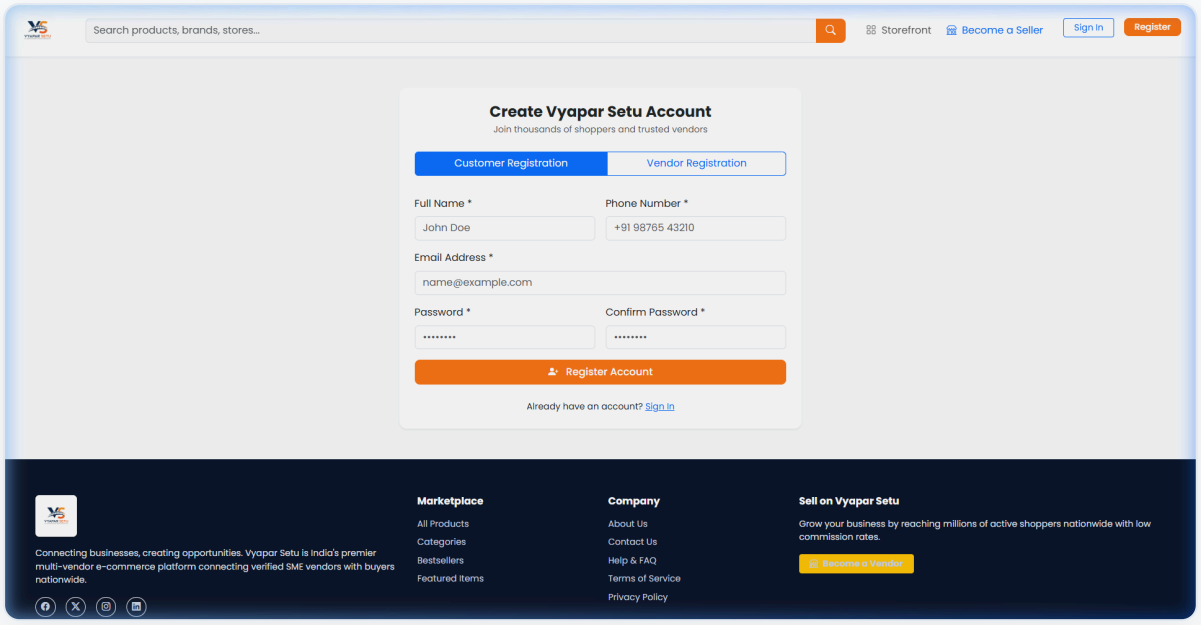


Figure 9.6 — Registration form with toggle between Customer and Vendor modes; vendor registration includes additional fields for store name, category, and GSTIN.

9.7 Customer Dashboard

Customer Dashboard — Rishiraj Sharma

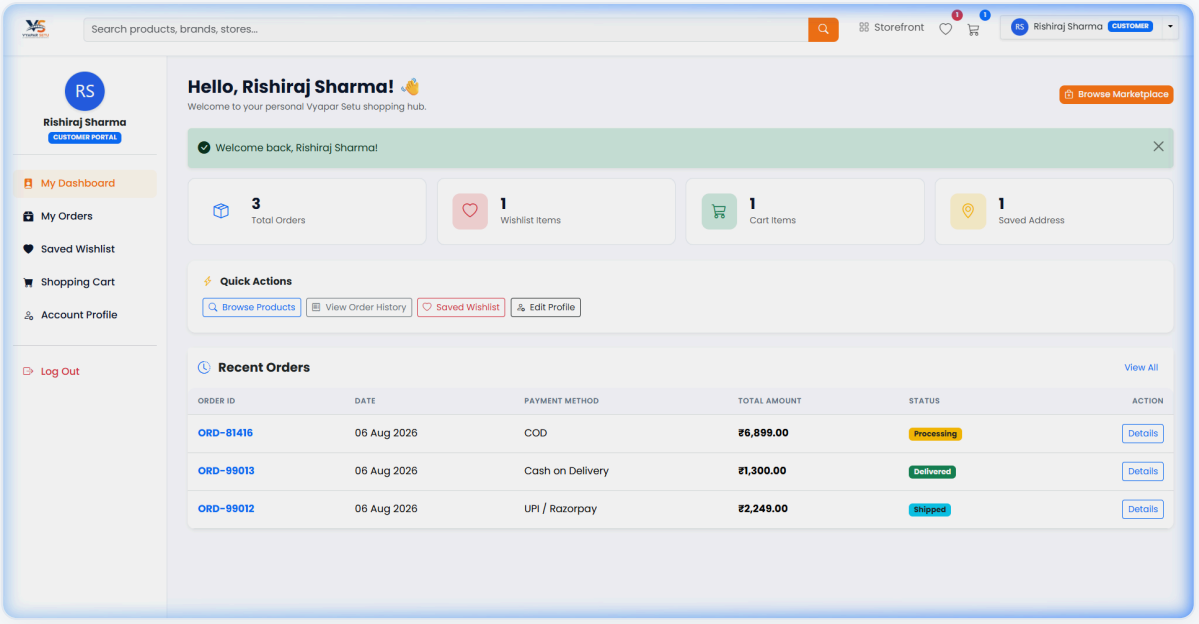


Figure 9.7 — Customer portal showing order metrics (Total Orders, Wishlist Items, Cart Items), quick action buttons, and recent orders table with status badges.

9.8 Shopping Cart

Shopping Cart — Order Summary

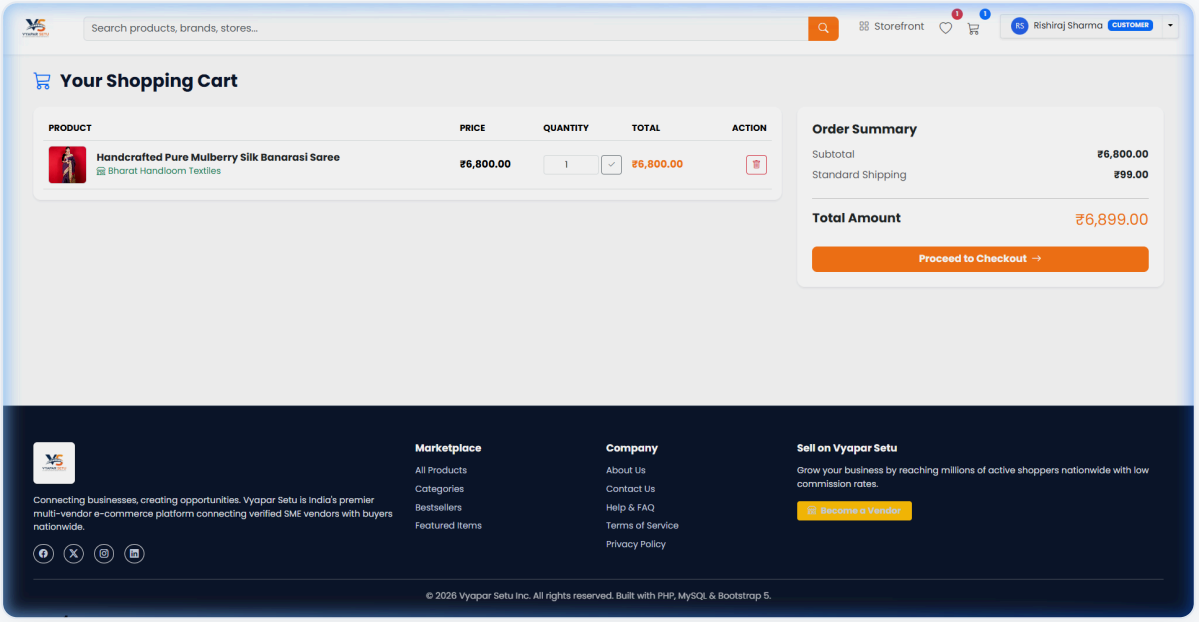


Figure 9.8 — Cart page with product details, quantity controls, vendor attribution, subtotal/shipping calculation, and checkout button.

9.9 Vendor Dashboard

Vendor Dashboard — TechCraft Electronics

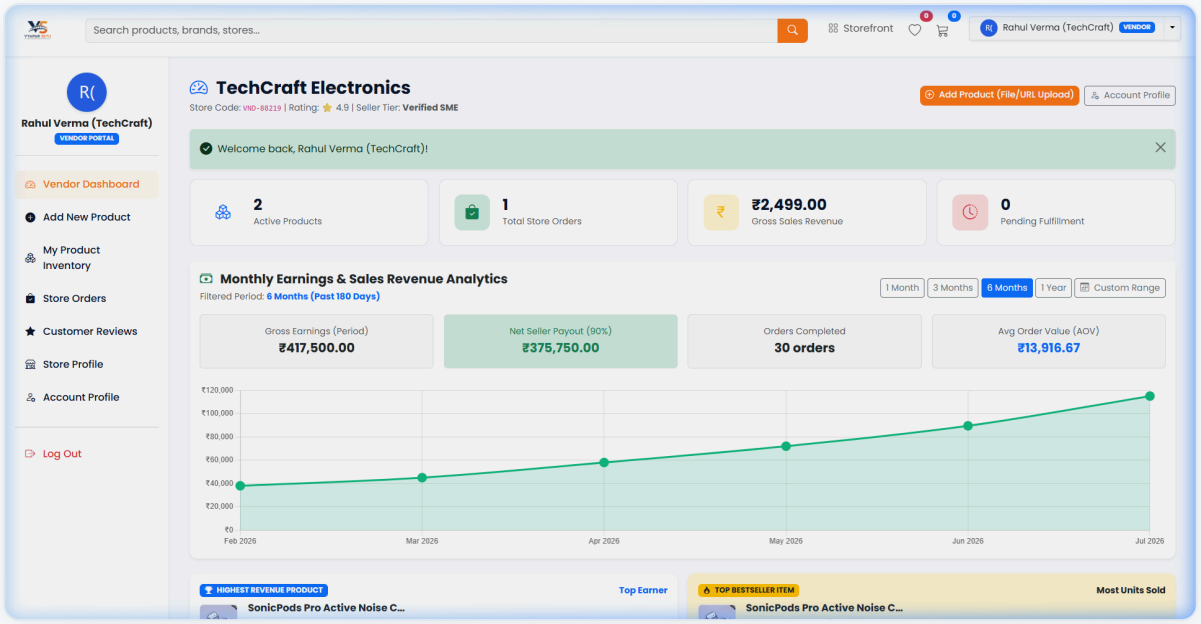


Figure 9.9 — Vendor portal showing store metrics (Active Products, Orders, Revenue), earnings analytics with Chart.js graphs, and top-performing products.

9.10 Admin Governance Dashboard

Admin Dashboard — Platform Governance

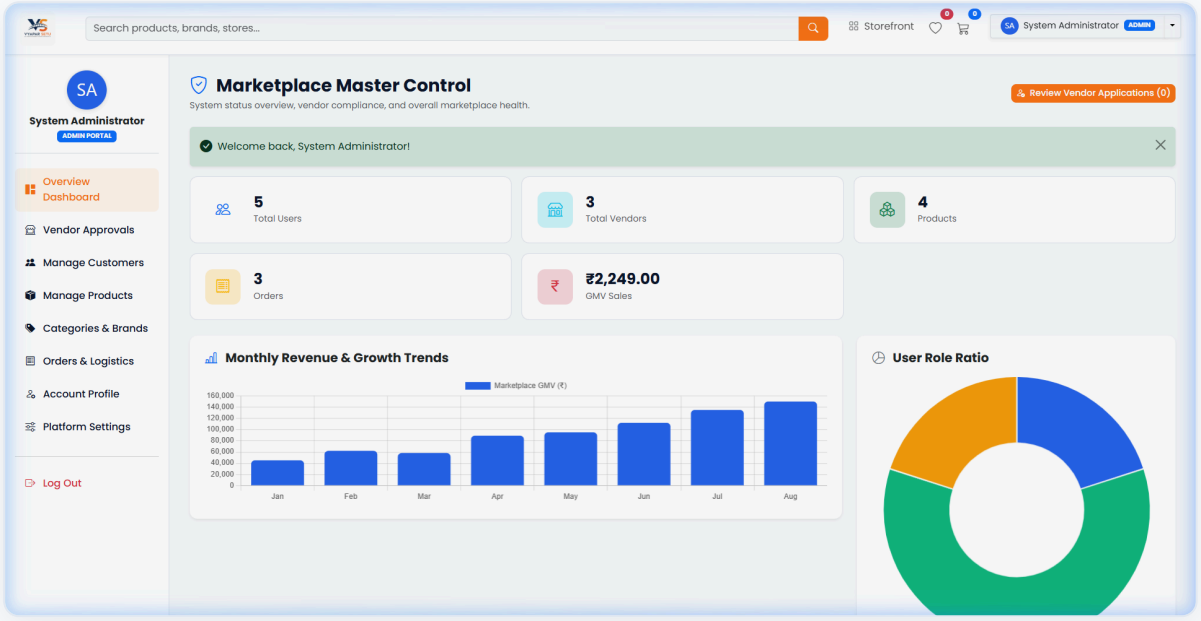


Figure 9.10 — Admin governance panel showing platform-wide statistics (Users, Vendors, Products, Orders, GMV), vendor approval queue, revenue trends, and user role distribution charts.

10 Advantages & Disadvantages

✅ Advantages

- **Multi-Vendor Support:** Multiple sellers can operate independently through a single platform, expanding product variety for customers.
- **Role-Based Access Control:** Three distinct user roles with dedicated dashboards ensure data isolation and proper authorization.
- **Zero-Configuration Deployment:** SQLite fallback eliminates the need for MySQL setup during development and demonstrations.
- **Responsive Design:** Bootstrap 5 ensures the platform works across desktops, tablets, and mobile devices.
- **Secure Architecture:** Prepared statements prevent SQL injection; bcrypt hashing protects passwords; session-based auth controls access.
- **AJAX-Powered Interactions:** Cart and wishlist operations work without page reloads, improving user experience.
- **Low Commission Model:** 10% default commission rate enables affordable entry for small vendors compared to major marketplaces.
- **Complete Analytics:** Both vendor and admin dashboards include revenue charts, sales metrics, and performance indicators.
- **Self-Hosted:** Platform can be deployed on any PHP-compatible server, giving full data ownership and control.
- **Open Source Stack:** Built entirely with free, open-source technologies (PHP,

❌ Disadvantages

- **No Payment Gateway Integration:** The platform lacks integration with real payment processors (Razorpay, Stripe, PayPal).
- **Limited Search:** Product search is basic text-matching without full-text search indexing or AI-powered recommendations.
- **No Real-Time Notifications:** Notifications are database-driven without WebSocket or push notification support.
- **Single-Server Architecture:** Not designed for horizontal scaling; may face performance issues under very high concurrent load.
- **No Email Verification:** Email verification is flagged in the database but actual email sending (SMTP) is not implemented.
- **No Multi-Language Support:** The platform is English-only; no i18n/l10n framework is integrated.
- **No Image CDN:** Product images use external URLs or local storage without CDN optimization for fast delivery.
- **Limited Reporting:** Financial reports and tax calculations are simplified; no GST invoice generation.
- **No Mobile App:** Only a web-based interface; no native Android/iOS companion applications.

MySQL, Bootstrap, Chart.js).

11 Analysis

11.1 System Architecture Analysis

Vyapar Setu follows a **monolithic server-rendered architecture** where all business logic, database operations, authentication, and HTML rendering happen within PHP scripts on the server. This approach provides simplicity of deployment and maintenance — the entire application can be deployed by copying files to a web server directory — but limits scalability compared to microservice or API-first architectures.

11.2 Database Design Analysis

The database schema consists of **16 well-normalized tables** covering the complete e-commerce domain. Key design decisions include:

- **Separation of Users and Vendors:** Vendor-specific data (store_name, GSTIN, commission_rate) is stored in a separate table linked via `user_id`, following the principle of data normalization.
- **Cart as a persistent entity:** The two-table cart design (cart + cart_items) allows future support for guest carts via session_id and logged-in carts via user_id.
- **Denormalized order items:** Order items store product_title and price at the time of purchase, ensuring historical accuracy even if the product is later modified or deleted.
- **Dual-database compatibility:** The PDO abstraction layer with SQLite fallback enables zero-configuration deployment, making the platform ideal for academic demonstrations.

11.3 Security Analysis

VULNERABILITY	MITIGATION	STATUS
SQL Injection	PDO prepared statements with parameterized queries	✅ Protected

VULNERABILITY	MITIGATION	STATUS
Cross-Site Scripting (XSS)	`htmlspecialchars()` via `sanitize()` on all output	✅ Protected
Password Storage	bcrypt hashing via `password_hash()`	✅ Protected
Session Hijacking	Session regeneration on login (recommended enhancement)	⚠️ Partial
CSRF Protection	Not implemented (recommended enhancement)	❌ Missing
File Upload Validation	MIME type checking, size limits, extension validation	✅ Protected

11.4 Performance Considerations

The platform performs efficiently for small to medium-scale deployments (up to ~10,000 products, ~1,000 concurrent users). For larger scale, the following optimizations would be recommended:

- Database indexing on frequently queried columns (category_id, vendor_id, status)
- Pagination for product listings (currently loads all matching products)
- Redis/Memcached caching for session storage and frequently accessed data
- CDN integration for static assets and product images
- Database connection pooling for high-concurrency environments

11.5 Comparison with Existing Platforms

FEATURE	VYAPAR SETU	AMAZON / FLIPKART	WOOCOMMERCE
Multi-Vendor	✅ Built-in	✅ Built-in	⚠️ Plugin required
Self-Hosted	✅ Full control	❌ SaaS-only	✅ WordPress-based
Commission Rate	10% (configurable)	15-30%	0% (self-hosted)
Setup Complexity	Low (XAMPP copy)	N/A (managed)	Medium (WordPress)
Vendor Dashboard	✅ Full analytics	✅ Seller Central	⚠️ Plugin dependent

FEATURE	VYAPAR SETU	AMAZON / FLIPKART	WOOCOMMERCE
Payment Gateway	❌ Simulated	✅ Integrated	✅ Multiple plugins
Technology	PHP + Bootstrap	Proprietary	PHP + WordPress

12 Conclusion

The Vyapar Setu multi-vendor marketplace platform has been successfully designed, developed, and tested as a comprehensive e-commerce solution that addresses the need for an affordable, self-hosted marketplace platform for Indian SMEs and independent artisans. The project demonstrates the practical application of core web development concepts including server-side programming with PHP, relational database design with MySQL/SQLite, responsive front-end development with Bootstrap 5, and AJAX-based interactive user interfaces.

The platform achieves all eight of its stated objectives: implementing a three-tier role-based access control system, building a complete product-to-order lifecycle, providing vendor self-service tools with analytics, and enabling administrative governance capabilities. The dual-database architecture (MySQL + SQLite fallback) ensures that the platform can be deployed and demonstrated without any external database configuration, making it particularly suitable for academic evaluation and rapid prototyping.

Key accomplishments include:

- A 16-table normalized database schema covering the complete e-commerce domain
- Three dedicated dashboards with role-specific features and metrics
- AJAX-powered cart and wishlist operations for seamless user experience
- Secure authentication with bcrypt password hashing and session management
- Vendor analytics with Chart.js-powered revenue and performance visualizations
- Complete order lifecycle from placement to tracking with status management

12.1 Future Scope

The Vyapar Setu platform can be enhanced in the following ways:

1. **Payment Gateway Integration:** Integrate Razorpay, Stripe, or PayPal for real payment processing with UPI, cards, and net banking support.
2. **AI-Powered Recommendations:** Implement collaborative filtering or content-based recommendation algorithms to suggest products based on browsing history.

3. **Real-Time Notifications:** Add WebSocket or Server-Sent Events (SSE) for instant order updates and chat functionality.
4. **Mobile Application:** Develop React Native or Flutter companion apps for Android and iOS platforms.
5. **Multi-Language Support:** Implement i18n framework for Hindi, Tamil, Bengali, and other regional language support.
6. **GST Invoice Generation:** Automated GST-compliant invoice generation for vendors and customers.
7. **Logistics Integration:** API integration with courier services (Shiprocket, Delhivery) for real-time tracking.
8. **CSRF Token Protection:** Add anti-CSRF tokens to all forms for enhanced security.

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Built with PHP 8.x · MySQL/SQLite · Bootstrap 5 · Chart.js · JavaScript